Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

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Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. We derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation includes posterior-matching metrics, predictive performance, calibration quality, mixing, and explicit computational cost accounting. Across these six benchmark families, multirate SVGD variants

improve robustness and quality-cost tradeoffs relative to vanilla SVGD. The strongest gains appear on stiff hierarchical, strongly anisotropic, and multimodal targets, where adaptive multirate SVGD is usually the strongest variant and fixed multirate SVGD provides a simpler robust alternative at lower cost.
Keywords: Stein variational gradient descent, Bayesian computation, multirate integration, particle-based variational inference, adaptive time stepping

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1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based view is central to particle-based variational inference [3]. Within this class, SVGD is a prominent deterministic representative, and recent work has examined acceleration, particle degeneracy, and kernel design in related Stein-particle methods [3, 4, 5]. It opens new avenues for improving sampling efficiency and robustness, especially in challenging inference settings.

A useful unifying perspective is to view interacting particle methods as discretizations of a particle flow (possibly with a stochastic component) that transports an initial ensemble toward the target distribution [6]. Here, we focus on Stein variational gradient descent (SVGD) [7] as a representative interacting particle method. SVGD constructs a velocity field for the particles based on the target density and a reproducing kernel Hilbert space (RKHS) structure, and updates the particles by discretizing this flow [7]. The velocity field naturally separates into an attractive drift, which moves particles toward high-density regions, and a repulsive interaction, which maintains particle diversity and support coverage of multiple modes.

This drift-repulsion structure creates the main numerical challenge studied in this paper. When particles crowd, repulsion can become stiff and destabilize updates. On ther hand, the drift can require finer resolution on anisotropic targets to follow the geometry of $\log p$ closely. A single step size must therefore balance competing requirements and can either over-damp repulsion, leading to loss of diversity and mode coverage, or under-resolve drift, leading to slow progress.

Moreover, repulsion typically requires kernel evaluation at each step, so increasing the stability by reducing the step-size can become computationally expensive. Finally, repulsive forces can weaken in high dimensions or under limited particle resolution, leading to particle collapse and poor support of multiple modes [4, 8]. In practice, the resulting single-step-size compromise is both statistically and computationally inefficient: it can degrade support coverage and predictive quality while still spending substantial kernel work to stabilize the wrong part of the update.

A common approach to solving systems with significantly different components is to use partitioned methods such as implicit-explicit (IMEX) [9, 10] and multirate integration methods [11]. In the context of ordinary differential equations (ODEs), multirate methods exploit a natural splitting of the dynamics into fast and slow components, allowing each to be integrated with a step size appropriate to its scale [12, 13, 14]. For example, in a split ODE of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent, smaller steps to the stiff or rapidly varying part, while updating the slow part less often. This approach improves both stability and efficiency in multi-scale dynamical systems, and has been successfully applied in a variety of scientific computing contexts [10, 15].

Motivated by advances in multirate time-integration for multi-scale dynamical systems—including MrGARK, IMEX, and related splitting schemes [12, 13, 14] we propose to integrate the drift and repulsion terms on separate time scales. This leads to a number of multirate SVGD variants, including symmetric splitting, fixed repulsion substepping, and adaptive error-controlled drift substepping.

To assess the effectiveness of these methods, we conduct a comprehensive empirical evaluation. We compare our multirate SVGD variants with established approaches from the literature across a range of benchmark problems, using a variety of task-appropriate metrics. Results highlight the strengths and limitations of each method under diverse geometric and statistical challenges.

Contributions

The main contributions of this work are:

- Multirate SVGD formulations that explicitly separate drift and repulsion and support symmetric splitting and fixed multirate schedules.
- An adaptive error-controlled multirate SVGD variant that selects substeps based on a local error estimator while keeping repulsion stable.
- A broad empirical study showing that multirate SVGD improves robustness and quality-cost tradeoffs compared to SVGD and representative chain methods, with the strongest gains on anisotropic, multimodal, and hierarchical targets.
- A benchmark suite spanning many different targets, equipped with appropriate diagnostics, early stopping, performance evaluation metrics for efficiency comparisons.

We present our work as follows. Section 2 presents the SVGD variants and the multirate construction. Section 3 describes the benchmarks, diagnostics, and experimental protocol. Section 4 reports the results. Section 5 summarizes the findings and discusses directions for future work.

2. Method

2.1. Background: SVGD as a particle flow

Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of particles at iteration t. SVGD evolves particles by following a deterministic interacting-particle flow whose velocity field is constrained to RKHS [7]. A first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h\,\phi(x_i^t), \tag{1}$$

where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N}\sum_{j=1}^{N}\Big[k(x_j, x)\,\nabla_{x_j}\log p(x_j) + \nabla_{x_j}k(x_j, x)\Big]. \tag{2}$$

The first term in eq. (2) is an *attractive* drift that pushes particles toward high-density regions via the score $\nabla \log p$. The second term is a *repulsive* interaction that discourages particle collapse. The multirate constructions below are kernel-agnostic: they apply once a kernel-induced attractive field and repulsive field have been specified.

2.2. Drift–repulsion splitting and multirate integration

The central numerical observation motivating this work is that the drift and repulsion components of eq. (2) can impose different stability constraints and can naturally evolve on different time scales. We therefore write the SVGD flow as a split system

$$\dot{x} = f_{\mathrm{rep}}(x) + f_{\mathrm{drift}}(x), \tag{3}$$

where, for particle i,

$$f_{\mathrm{drift}}(x_i) := \frac{1}{N}\sum_{j=1}^{N} k(x_j, x_i)\,\nabla_{x_j}\log p(x_j), \tag{4}$$

$$f_{\mathrm{rep}}(x_i) := \frac{1}{N}\sum_{j=1}^{N} \nabla_{x_j}k(x_j, x_i). \tag{5}$$

We next define multirate discretizations that apply different substepping strategies to f_{drift} and f_{rep} while keeping a common macro-step size h.

Strang-split SVGD. Define the forward Euler update that advance the split system by a step size τ:

$$\Psi^{\tau}_{\text{rep}}(x) = x + \tau f_{\text{rep}}(x), \qquad \Psi^{\tau}_{\text{drift}}(x) = x + \tau f_{\text{drift}}(x).$$

Strang splitting [16] is a classical second-order method that constructs a macro-step of size h by composing such substeps. In our setting, one macro step takes the form

$$x^{t+1} = \Psi^{h/2}_{\text{rep}} \circ \Psi^{h}_{\text{drift}} \circ \Psi^{h/2}_{\text{rep}}(x^t), \tag{6}$$

which alternates half a repulsion step, a full drift step, and another half repulsion step. In practice, the half-steps can be implemented as $\frac{h}{2m}$ smaller substeps for stability.

Fixed multirate SVGD (MR-SVGD). Fixed multirate integrators provide a principled way to allocate different step sizes to fast and slow components of a split system [11, 12, 13]. In our SVGD setting, however, higher-order multirate schemes are often not cost-effective: their additional stages translate into multiple evaluations of the drift and repulsion fields per macro step, and each such evaluation is expensive due to gradient computations and, in particular, $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by finite-particle effects and by choices such as bandwidth selection and early stopping, which can mute the benefits of increasing time-integration order at a fixed computational budget. We therefore utilize a first-order multirate construction using Euler updates that preserves a macro step of size h for the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi^{h}_{\text{drift}} \circ \left(\Psi^{h/m}_{\text{rep}}\right)^m (x^t). \tag{7}$$

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^{N}$, step size h, repulsion substeps m, kernel k

```
x ← {x_i}
for s = 1 to m do
    compute repulsion f_rep(x) using eq. (5)
    x ← x + (h/m) f_rep(x)
end for
compute drift f_drift(x) using eq. (4)
x ← x + h f_drift(x)
return x
```

This retains a coarse drift step while stabilizing the repulsive interaction when particles crowd, at the cost of additional kernel evaluations. The corresponding implementation-level procedure is summarized in algorithm 1.

Adaptive multirate SVGD (Adapt-MR-SVGD). Although fixed-m schedules are simple and robust, they impose a uniform drift resolution along the entire trajectory. As a result, they may spend unnecessary computational effort in benign regions while providing insufficient resolution in locally stiff regimes. To address this imbalance, we choose the number of drift microsteps m adaptively at each macro step, while retaining one repulsion update per macro step. This follows the standard local-error-control perspective used in ODE integrators, where local error estimates are used to adjust resolution [17, Ch. II]. After the repulsion update, denote the state by $x = \Psi_{\mathrm{rep}}^{h}(x^t)$. We estimate the local drift-discretization error by comparing one drift step of size h

with two consecutive drift half-steps of size $h/2$,

$$x_{\text{full}} = x + h\, f_{\text{drift}}(x), \tag{8}$$

$$x_{\text{half}} = x + \tfrac{h}{2}\, f_{\text{drift}}(x), \tag{9}$$

$$\tilde{x}_{\text{full}} = x_{\text{half}} + \tfrac{h}{2}\, f_{\text{drift}}(x_{\text{half}}). \tag{10}$$

We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\text{full}} - x_{\text{full}}\|_2}{\|x\|_2 + \epsilon}, \tag{11}$$

which serves as a dimensionless local error indicator for the drift update. $\epsilon > 0$ is a small constant to avoid division by zero. For a first-order drift integrator, this local error behaves as [17, Ch. II.4]:

$$\rho(h) = Ch^2 + \mathcal{O}(h^3),$$

where C is the principal error constant dependent on f and its derivatives. If the drift step is replaced by m microsteps of size h/m, then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

Imposing the target local error tolerance $\rho_m \approx \text{Tol}$ gives $m \approx \sqrt{\rho/\text{Tol}}$. We therefore use the clipped integer controller

$$m \leftarrow \text{clip}\Big(\Big\lceil \sqrt{\rho/\text{Tol}} \Big\rceil, m_{\min}, m_{\max}\Big), \tag{12}$$

which increases m when $\rho > \text{Tol}$ and decreases it when $\rho \leq \text{Tol}$, while enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update is then

$$x^{t+1} = \Big(\Psi_{\text{drift}}^{h/m}\Big)^m \circ \Psi_{\text{rep}}^{h}(x^t), \qquad m = m(x^t, h). \tag{13}$$

This strategy increases drift resolution only when the local error indicator suggests stiffness or rapid variation in the drift component. Algorithm 2 provides the complete algorithm for this method.

Remark 1. In practice, if $m \leq 2$ we can reuse the already computed, locally-higher-order solution $\tilde{x}_{\text{full}}$ to avoid extra drift evaluations.

3. Experimental Setup

We evaluate multirate SVGD variants on six benchmark groups: a 50D Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark. The comparison focuses on both statistical quality and computational cost. This section defines the shared diagnostics and protocol; benchmark-specific details are introduced where needed below.

3.1. Evaluation Metrics

Evaluation combines distributional fidelity, predictive performance, mixing, and explicit computational cost accounting. Metrics used across multiple benchmarks are defined here once; benchmark-specific diagnostics are introduced locally.

Moment fidelity. When reference moments are available, we report $e_\mu = \|\hat{\mu} - \mu_{\text{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\text{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i$ and $\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical in the Gaussian and synthetic-target settings.

Stein discrepancy. For distributional fidelity, we use kernel Stein discrepancy (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot, \cdot)$ [18, 19]. The corresponding Stein kernel is

$$\begin{aligned}\kappa_p(x, x') = s_p(x)^\top s_p(x')\, k_h(x, x') + s_p(x)^\top \nabla_{x'} k_h(x, x') \\ + s_p(x')^\top \nabla_x k_h(x, x') + \mathrm{tr}(\nabla_x \nabla_{x'} k_h(x, x'))\,,\end{aligned} \tag{14}$$

and the empirical discrepancy is

$$\mathrm{KSD}(X; p) = \left(\frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \kappa_p(x_i, x_j) \right)^{1/2}. \tag{15}$$

KSD measures how far the empirical particle distribution is from satisfying the target's Stein identity. Intuitively, it acts as a global goodness-of-fit metric: it is small when the particles collectively match the target law, and large when they exhibit systematic errors such as bias, incorrect spread, or missed modes.

Mean log-density fit. We complement KSD with the empirical mean log-density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^{N} \log p(x_i), \tag{16}$$

which measures how strongly the particles concentrate in high-density regions of the target. Unlike KSD, this is a pointwise criterion: it rewards samples for lying in locally plausible regions, but by itself it does not certify global distributional agreement. Reporting eqs. (15) and (16) therefore separates global distributional fidelity from local density concentration.

Predictive performance and calibration. For the binary-classification experiments (UCI and BNN tasks), we report accuracy, negative log-likelihood, and expected calibration error [1],

$$\mathrm{NLL} = -\frac{1}{M}\sum_{m=1}^{M}\Big[y_m \log \hat{p}_m + (1-y_m)\log(1-\hat{p}_m)\Big], \tag{17}$$

$$\mathrm{ECE} = \sum_{b=1}^{B}\frac{|\mathcal{B}_b|}{M}\Big|\mathrm{acc}(\mathcal{B}_b) - \mathrm{conf}(\mathcal{B}_b)\Big|. \tag{18}$$

In all predictive experiments, ECE is computed with $B = 10$ equal-width bins on $[0, 1]$. Together, these metrics capture thresholded classification performance, probabilistic predictive fit, and calibration quality.

Mixing efficiency. To quantify dependence, we report ESS as a supplemental univariate mixing diagnostic, computed from the first coordinate using a standard autocorrelation-based estimator with initial-positive truncation [20]. For particle methods at a checkpoint, the diagnostic sequence is formed from the first coordinate across particles; for single-chain baselines, it is formed from a rolling window of recent iterates. Because this quantity is one-dimensional, we do not interpret ESS as a full multivariate efficiency measure but as a complementary statistic.

Cost accounting. Alongside wall-clock time, we track gradient and kernel evaluation counts to support hardware-agnostic efficiency comparisons across methods with different workloads.

Remark 2. For the common metrics above, lower is better for e_μ, e_Σ, KSD, NLL, ECE, wall-clock time, and gradient- and kernel-evaluation counts, whereas

[1] Accuracy as the fraction of correctly classified examples under a 0.5 threshold on the posterior predictive probability. $\mathrm{acc}(\mathcal{B}_b)$ is the empirical accuracy in bin $\mathcal{B}_b$ under the same threshold, and $\mathrm{conf}(\mathcal{B}_b)$ is the mean predictive probability in that bin.

higher is better for ESS, accuracy, and mean log-probability.

3.2. Experimental Protocol

Across benchmarks, we compare the same method family whenever applicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain baselines SGLD and an SGHMC-style method. The SVGD-family updates are those defined in section 2, namely eqs. (1), (6), (7) and (13).

Single-chain baselines use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta}\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0, I), \tag{19}$$

for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta}\, \xi_t, \qquad x_{t+1} = x_t + v_{t+1}, \tag{20}$$

for an SGHMC-style baseline [2]. Here η is the step size, α is a friction or noise parameter, and $\beta \in (0, 1)$ is the momentum decay.

Particle configuration and kernel choice. For particle methods, we use $N = 128$ particles; for single-chain methods, we compute metrics from a rolling window of recent iterates to obtain comparable finite-sample summaries. Unless stated otherwise, particle methods use a Gaussian RBF kernel with bandwidth selected by the median heuristic and a benchmark-specific scaling factor. For Mix8, we instead use a shared multiscale RBF kernel with bandwidth factors $\{0.5, 1, 2\}$ to better probe local spacing and longer-range interactions across modes.

Implementation safeguards. To improve numerical robustness, the multirate implementations clip raw score components to the interval $[-50, 50]$ before assembling the update. Benchmarks with pronounced instability additionally use a short non-finite fail-fast guard, which we note locally when it is active.

Checkpointing and model selection. All benchmarks use checkpointed evaluation at benchmark-specific intervals and a patience-based early-stopping rule triggered by persistent degradation of a designated monitor metric. Pure sampling benchmarks use KSD as the monitor (50D Gaussian, 2D synthetic targets, and Mixture2D), whereas predictive benchmarks use held-out NLL (UCI logistic regression, BNN, and HLR). For each run, we retain the best finite checkpoint under that monitor before stopping. Accuracy and ECE are reported for predictive tasks but are never used as stopping criteria.

4. Results

In the following, we introduce each benchmark, explain its relevance to particle-based sampling, and report the corresponding empirical results. The benchmark suite consists of a 50D Gaussian target, 2D synthetic targets, a multimodal Gaussian mixture (Mix8), UCI logistic regression, a Bayesian neural network, and large-scale hierarchical logistic regression.

4.1. 50D Gaussian

This benchmark tests the stability of sampling in a high-dimensional setting with anisotropic covariance structure. We run 1,000 sampling iterations with checkpointing every 50 steps and a 5-seed ablation. Reported metrics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and time-to

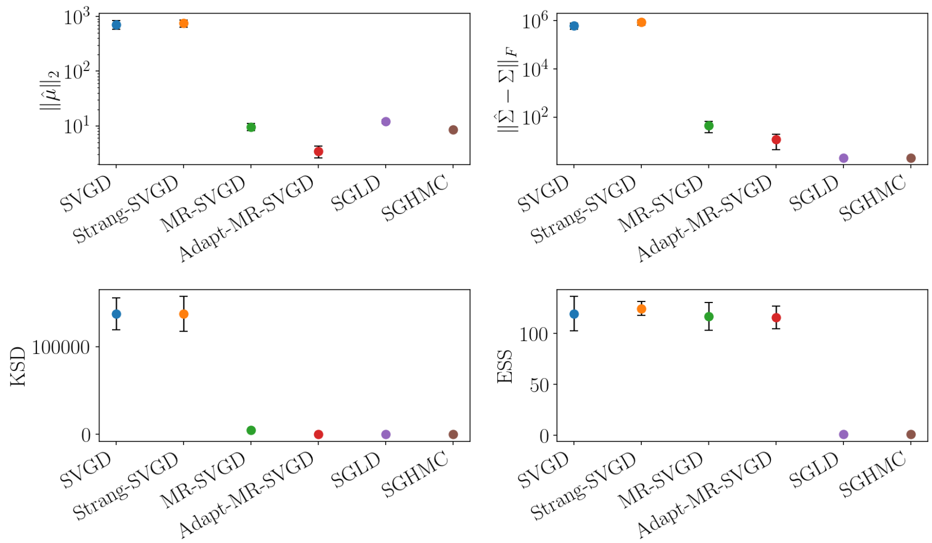


Figure 1: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively; markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS. This figure highlights robustness differences under strong anisotropy: Adapt-MR-SVGD is the only particle variant that keeps both moment errors and KSD under control.

best-checkpoint.

Among the methods, fig. 1 shows that Adapt-MR-SVGD is the only variant that keeps both moment errors and KSD under control in this anisotropic Gaussian setting: it substantially improves on MR-SVGD and avoids the severe degradation exhibited by vanilla SVGD and Strang-SVGD. The chain baselines remain competitive on some final-checkpoint fidelity metrics, but they do so with ESS near 4, far below the particle methods. Within the particle family, fig. 2 shows that MR-SVGD offers a cheaper but less accurate baseline, whereas Adapt-MR-SVGD achieves the strongest quality at a

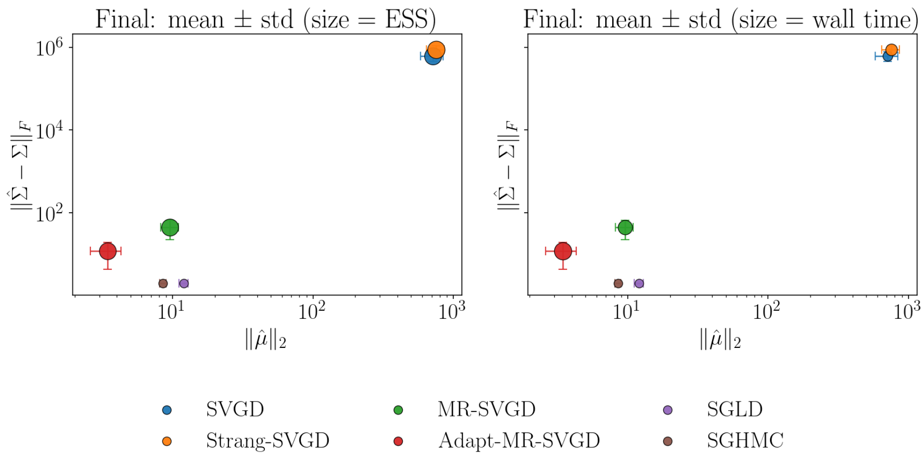


Figure 2: 50D Gaussian: final mean ± std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

higher computational cost.

4.2. 2D synthetic targets

In this benchmark, we evaluate a suite of 2D targets with varying geometry and multimodality, including banana, ring, squiggly, two-moons, and funnel. To contextualize the geometry of these targets, fig. 3 shows short-run visualization panels for representative cases. Results are summarized over five seeds, each with 1,000 iterations and checkpointing every 20 steps. Here KSD serves both as the stop monitor and the primary quality metric. We add a safety guard against non-finite KSD values in the stopping policy, because several targets can produce unstable particle trajectories. The main comparison in table 1 focuses on KSD, mean log probability, gradient-evaluation cost, and wall time. The panels in fig. 3 are qualitative only and are included to illustrate target geometry and short-run particle behavior.

Table 1 shows a sumamry of the metrics for the 2D suite. Banana is effectively a tie: all four particle methods reach nearly identical best KSD values, with only minor differences in mean log probability and cost. The other targets separate more sharply. On ring, squiggly, and two moons, Adapt-MR-SVGD is the only method that stays near the target while keeping KSD below 1, whereas the other particle methods remain orders of magnitude worse in both KSD and mean log probability. MR-SVGD is often cheaper in gradient-evaluation count on those targets, but it pays for that reduction with substantially weaker fidelity.

Funnel is the main exception. There MR-SVGD attains the best median KSD, but Adapt-MR-SVGD has much better median mean log probability while also using fewer gradient evaluations and less wall time. Taken together, these results show that adaptive multirate splitting is not uniformly best on every benign 2D target, but it is the only particle variant that combines strong distributional fidelity with competitive cost across the harder geometries.

4.3. Mixture2D (mix8)

Mix8 is an equal-weight eight-component Gaussian mixture with modes arranged uniformly on a ring of radius 4. All methods start from the same small Gaussian centered at the origin, away from those modes, and then run for a fixed budget of 1,000 iterations with checkpointing every 20 steps across 5 seeds. For this benchmark we use a shared multiscale RBF kernel with bandwidth factors $\{0.5, 1, 2\}$ for all particle methods so that local spacing and longer-range repulsion are both represented. This also demonstrates that the multirate formulation is agnostic to kernel choice: here we use a richer kernel than a single-bandwidth RBF to capture both local and longer-range interactions. Prior SVGD work has also considered broader kernel families beyond a single scalar RBF [5, 21].

Let $a(x_i)$ denote the nearest mixture center for particle x_i, and let

$$p_k = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{a(x_i) = k\}$$

be the empirical mass of mode k. We define mode coverage as the fraction of modes with $p_k \geq 0.05$, normalized mode entropy as $-\sum_{k=1}^{K} p_k \log p_k / \log K$, and mode imbalance as $\mathrm{std}(p_1, \ldots, p_K)$. Thus, higher coverage and entropy indicate broader exploration, while lower imbalance indicates more even mass allocation across discovered modes. The primary metrics are therefore mode coverage, normalized mode entropy, and mode imbalance. KSD and mean log probability are retained as secondary diagnostics. The fixed-budget results are summarized in fig. 4, which reports the mean final metrics across seeds together with one standard deviation.

Figure 4 shows that Adapt-MR-SVGD gives the strongest fixed-budget multimodal result. It attains the highest mean final coverage and entropy across seeds (0.500 and 0.544, respectively) while keeping KSD at 0.360, far below the single-rate particle baselines. Vanilla SVGD and Strang-SVGD spread farther around the ring than MR-SVGD, but that extra spread is unstable, with final KSD in the 10^3 range. Fixed MR-SVGD is much more stable than those baselines, yet it still underexplores the ring, reaching only 0.250 coverage and 0.238 entropy.

The imbalance panel further shows that Adapt-MR-SVGD distributes mass more evenly across the modes it discovers. By contrast, SGLD and SGHMC remain near a single mode, with coverage fixed at 0.125 and zero entropy, so their small KSD values do not indicate successful multimodal exploration.

4.4. UCI logistic regression

We benchmark Bayesian logistic regression on breast_cancer, ionosphere, spambase, and a5a, using a Gaussian prior on weights (as implemented in the benchmark script) and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, and mean log probability; KSD is optional and disabled by default in this benchmark configuration. Results are summarized with per-dataset panels in fig. 5 and a grouped best-NLL summary table (particle methods only) in table 2; both use the best finite-NLL checkpoint selected under the global predictive stopping policy [22].

UCI logistic regression still yields the weakest particle-method separation in the paper, but the refreshed summary in table 2 is more favorable to Adapt-MR-SVGD than before. Among particle methods, it now achieves the best NLL on breast_cancer, ionosphere, and a5a, while vanilla SVGD is best on spambase. The per-dataset panels in fig. 5 extend this comparison to the chain baselines and show that SGHMC, and on ionosphere also SGLD, can still be competitive or superior in NLL at much lower wall time, but with ESS far below the particle methods. Accuracy remains tightly clustered on these datasets, so NLL is the more discriminative predictive metric. Overall, the rerun leaves the qualitative conclusion unchanged: adaptive multirate splitting is competitive on these relatively benign predictive posteriors, but its clearest gains appear on the more anisotropic, multimodal, and hierarchical targets.

4.5. Bayesian neural network

We evaluate a one-hidden-layer BNN (hidden width 32) on spambase and a5a, using a Gaussian prior, a fixed split seed, and 5 sampler seeds per dataset. For each dataset, we carve a validation split from the training data, select checkpoints by validation NLL under the global predictive stopping policy, and report test accuracy, test NLL, test ECE, ESS, and mean log probability at the selected checkpoint. KSD is optional and disabled by default in this configuration.

The two BNN datasets give a more informative but less uniform predictive story than the earlier small-data configuration. On spambase, Adapt-MR-SVGD attains the best overall test NLL (0.174) and the highest mean test accuracy (0.940), with the remaining particle methods clustered together around 0.187. On a5a, the ranking shifts: MR-SVGD attains the best overall test NLL (0.334) and the highest mean test accuracy (0.844), Adapt-MR-SVGD remains the second-best particle method (0.347), and the single-rate particle variants are again worse. The chain baselines are competitive in isolated cases—SGHMC on spambase and SGLD on a5a—but the multirate particle methods form the strongest pair overall across the two datasets. This makes the BNN benchmark corroborative rather than primary evidence: multirate structure improves predictive quality over the single-rate particle baselines, but the preferred multirate variant depends on the dataset.

4.6. Large-scale hierarchical logistic regression

To stress-test multirate behavior in a high-dimensional anisotropic posterior, we use a large-scale hierarchical logistic regression benchmark with grouped random effects. The model combines high parameter dimension with funnel-like geometry induced by a global scale parameter, making it a natural benchmark for joint quality-cost assessment of SVGD variants.

Problem definition. Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in \mathbb{R}^p$, and group indices $g_i \in \{1, \ldots, G\}$, we define

$$\begin{gathered} y_i \mid x_i, g_i, \beta, \alpha, u \sim \text{Bernoulli}\big(\sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\,, \\ i = 1, \ldots, n, \end{gathered} \tag{21}$$

where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for group effects,

$$\begin{aligned} \beta &\sim \mathcal{N}(0, \sigma_\beta^2 I_p), \quad && \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \\ u_j &= \tau z_j, && z_j \sim \mathcal{N}(0, 1), \\ \log \tau &\sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \end{aligned} \tag{22}$$

for $j = 1, \ldots, G$. The posterior (up to normalization) is

$$\begin{aligned} \pi(\theta) \propto & \left[\prod_{i=1}^{n} \text{Bernoulli}\big(y_i; \sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\right] \\ & \times p(\beta)\, p(\alpha)\, p(z)\, p(\log \tau), \\ \theta = & (\beta, \alpha, z, \log \tau). \end{aligned} \tag{23}$$

Protocol. For the long-tail grouped setting, we run $n = 10^6$ observations with $p = 300$ features and $G = 50{,}000$ groups. Each method runs up to 1,000 iterations with checkpointing every 20 iterations and 5 random seeds; particle methods use 32 particles, and single-chain baselines use a rolling-window evaluation set. This benchmark follows the global predictive NLL-based stopping policy and adds a two-checkpoint non-finite fail-fast guard because unstable methods can quickly leave the finite-NLL regime. ECE is reported as a metric but is not used as a stopping trigger.

Result summary. table 4 shows that Adapt-MR-SVGD achieves the best mean best-NLL while remaining finite across all seeds. MR-SVGD is a close second in NLL and reaches its best checkpoint faster on average (92.5 s versus 123.3 s). SVGD/Strang-SVGD and SGHMC show frequent non-finite behavior in this regime, while SGLD remains finite more often but with substantially worse NLL.

Uniform-group sensitivity check. table 5 provides a compact robustness check under uniform group assignments. Adapt-MR-SVGD remains best in mean best-NLL with full finite-seed coverage (5/5), and MR-SVGD remains close while being fastest among methods with full finite coverage. This supports the same ordering seen in the long-tail regime while keeping the long-tail setting as the primary stress test for our main claim.

5. Conclusions

This work studies a central numerical weakness of SVGD: a single global step size must simultaneously resolve attractive drift and diversity-preserving repulsion even when these components evolve on different effective time scales. We address this mismatch by recasting SVGD as a split interacting-particle flow and integrating the drift and repulsion terms with multirate updates.

Across a broad benchmark suite, the main empirical pattern is that multirate SVGD improves robustness and quality-cost tradeoffs relative to vanilla SVGD, with the strongest gains on anisotropic, multimodal, and hierarchical targets. Fixed multirate schedules provide a simpler robustness baseline, while adaptive drift resolution offers the strongest performance when local stiffness changes along the trajectory. At the same time, no single variant dominates every metric and every regime, reinforcing the importance of evaluating quality and cost jointly rather than through a single summary statistic.

Several limitations remain. Current adaptivity is centered on drift resolution, while repulsion frequency is still controlled by simple schedules; ESS remains a univariate proxy; and dense kernel interactions continue to constrain scaling at large particle counts. These observations suggest clear next steps: tighter adaptive controllers that coordinate drift and repulsion updates, approximate or structured kernel computations for large-N regimes, and stronger theoretical links between multirate stability analysis and particle-based Bayesian inference guarantees. Together, these directions can move multirate SVGD from a robust empirical strategy toward a more principled and scalable foundation for modern Bayesian computation.

Data availability

Code, experiment scripts, and manuscript sources for this study are maintained in a public GitHub repository.[2] Public benchmark inputs used in the predictive experiments are derived from standard UCI datasets; all other benchmark targets are synthetic and are generated directly by the repository code. Generated figures, metrics, and supplementary animation artifacts are available in the project repository for reference and reproduction.

[2] `https://github.com/csml-beach/multirate-sampling`

Declaration of generative AI and AI-assisted technologies

The jax codebase for this project was developed in part with the assistance of OpenAI Codex. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

[1] M. Welling, Y. W. Teh, Bayesian learning via stochastic gradient langevin dynamics, in: International Conference on Machine Learning, 2011.

[2] T. Chen, E. Fox, C. Guestrin, Stochastic gradient hamiltonian monte carlo, in: International Conference on Machine Learning, 2014.

[3] C. Liu, J. Zhuo, P. Cheng, R. Zhang, J. Zhu, Understanding and accelerating particle-based variational inference, in: K. Chaudhuri, R. Salakhutdinov (Eds.), Proceedings of the 36th International Conference on Machine Learning, Vol. 97 of Proceedings of Machine Learning Research, PMLR, 2019, pp. 4082–4092.
URL `https://proceedings.mlr.press/v97/liu19i.html`

[4] J. Zhuo, C. Liu, J. Shi, J. Zhu, N. Chen, B. Zhang, Message passing stein variational gradient descent, in: J. Dy, A. Krause (Eds.), Proceedings of the 35th International Conference on Machine Learning, Vol. 80 of Proceedings of Machine Learning Research, PMLR, 2018, pp. 6018–6027.
URL `https://proceedings.mlr.press/v80/zhuo18a.html`

[5] D. Wang, Z. Tang, C. Bajaj, Q. Liu, Stein variational gradient descent with matrix-valued kernels, in: Advances in Neural Information Processing Systems, 2019.

[6] A. N. Subrahmanya, A. A. Popov, A. Sandu, Ensemble variational fokker-planck methods for data assimilation, Journal of Computational Physics 523 (2025) 113681. doi:https://doi.org/10.1016/j.jcp.2024.113681.

[7] Q. Liu, D. Wang, Stein variational gradient descent: A general purpose bayesian inference algorithm, in: Advances in Neural Information Processing Systems, 2016.

[8] J. Zhang, R. Zhang, L. Carin, C. Chen, Stochastic particle-optimization sampling and the non-asymptotic convergence theory, in: S. Chiappa, R. Calandra (Eds.), Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, Vol. 108 of Proceedings of Machine Learning Research, PMLR, 2020, pp. 1877–1887.
URL `https://proceedings.mlr.press/v108/zhang20d.html`

[9] L. Pareschi, G. Russo, Implicit-explicit Runge-Kutta schemes for stiff systems of differential equations, in: Recent trends in numerical analysis, Nova Science Publishers, Inc., 2000, pp. 269–288.

[10] A. Sarshar, S. Roberts, A. Sandu, Parallel implicit-explicit general linear methods, Communications on Applied Mathematics and Computation 3 (2021) 649–669. doi:10.1007/s42967-020-00083-5.

[11] M. Günther, A. Sandu, Multirate generalized additive Runge-Kutta methods, Numerische Mathematik 133 (3) (2016) 497–524. doi:10.1007/s00211-015-0756-z.

[12] A. Sarshar, S. Roberts, A. Sandu, Design of high-order decoupled multirate gark schemes, SIAM Journal on Scientific Computing 41 (2) (2019) A816–A847. doi:10.1137/18M1182875.

[13] A. Sandu, M. Günther, S. Roberts, A. Sarshar, Implicit multirate gark methods, Journal of Scientific Computing 87 (2021) 1–32. doi:10.1007/s10915-020-01400-z.

[14] A. Sarshar, S. Roberts, A. Sandu, Alternating directions implicit integration in a general linear method framework, Journal of Computational and Applied Mathematics 387 (2021) 112496. doi:10.1016/j.cam.2019.112496.

[15] A. Sarshar, S. Roberts, A. Sandu, A fast time-stepping strategy for dynamical systems equipped with a surrogate model, SIAM Journal on Scientific Computing 44 (3) (2022) A1405–A1430. doi:10.1137/20M1386281.

[16] G. Strang, On the construction and comparison of difference schemes, SIAM Journal on Numerical Analysis 5 (3) (1968) 506–517.

[17] E. Hairer, S. Norsett, G. Wanner, Solving ordinary differential equations I: Nonstiff problems, no. 8 in Springer Series in Computational Mathematics, Springer-Verlag Berlin Heidelberg, 1993. doi:10.1007/978-3-540-78862-1.

[18] Q. Liu, J. Lee, M. Jordan, A kernelized stein discrepancy for goodness-of-fit tests, in: International Conference on Machine Learning, 2016.

[19] J. Gorham, L. Mackey, Measuring sample quality with kernels, in: D. Precup, Y. W. Teh (Eds.), Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of Proceedings of Machine Learning Research, PMLR, 2017, pp. 1292–1301.
URL https://proceedings.mlr.press/v70/gorham17a.html

[20] C. J. Geyer, Practical markov chain monte carlo, Statistical Science 7 (4) (1992) 473–483. doi:10.1214/ss/1177011137.

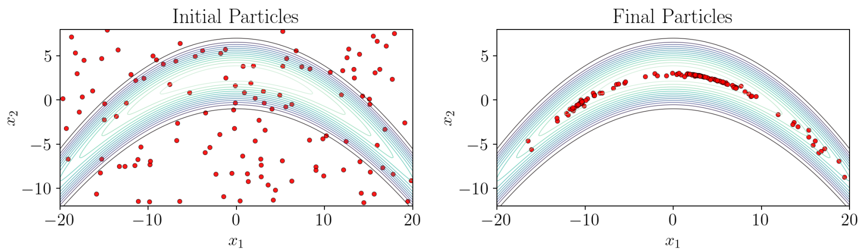
[21] A. S. Stordal, R. J. Moraes, P. N. Raanes, G. Evensen, p-kernel stein variational gradient descent for data assimilation and history matching, Mathematical Geosciences 53 (2021) 375–393. doi:10.1007/s11004-021-09937-x.

[22] D. Dua, C. Graff, Uci machine learning repository, https://archive.ics.uci.edu/ml, accessed 2025-01-01 (2019).

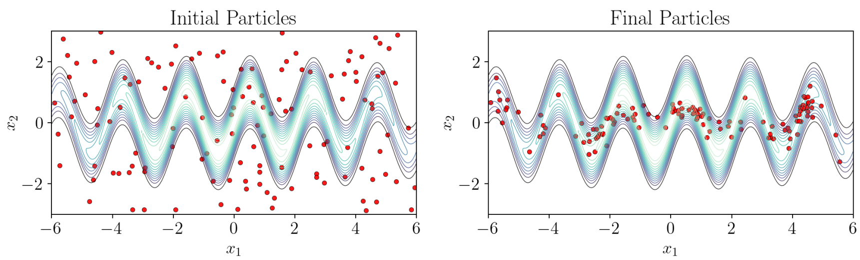
Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h, bounds $m_{\min}, m_{\max}$, tolerance Tol

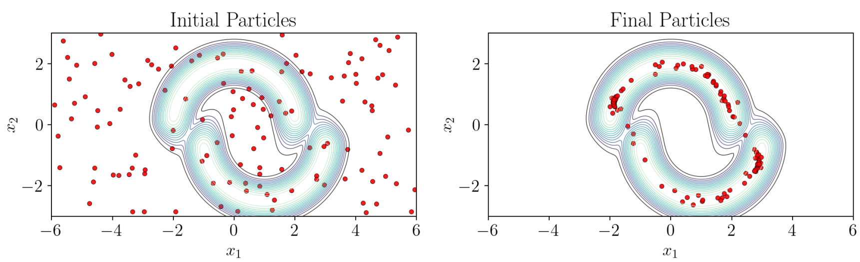
$x \leftarrow \{x_i\}$
compute repulsion $f_{\text{rep}}(x)$ using eq. (5)
$x \leftarrow x + h\, f_{\text{rep}}(x)$
compute drift $f_{\text{drift}}(x)$ using eq. (4)
$x_{\text{full}} \leftarrow x + h\, f_{\text{drift}}(x)$
$x_{\text{half}} \leftarrow x + (h/2)\, f_{\text{drift}}(x)$
compute drift $f_{\text{drift}}(x_{\text{half}})$
$\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2)\, f_{\text{drift}}(x_{\text{half}})$
estimate relative error $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\| / (\|x\| + \epsilon)$
$m \leftarrow \text{clip}(\lceil \sqrt{\rho/\text{Tol}} \rceil, m_{\min}, m_{\max})$
if $m \leq 2$ **then**
    $x \leftarrow \tilde{x}_{\text{full}}$ ▷ reuse current predictor if accurate
else
    for $s = 1$ to m **do**
        compute drift $f_{\text{drift}}(x)$
        $x \leftarrow x + (h/m)\, f_{\text{drift}}(x)$
    end for
end if
return x



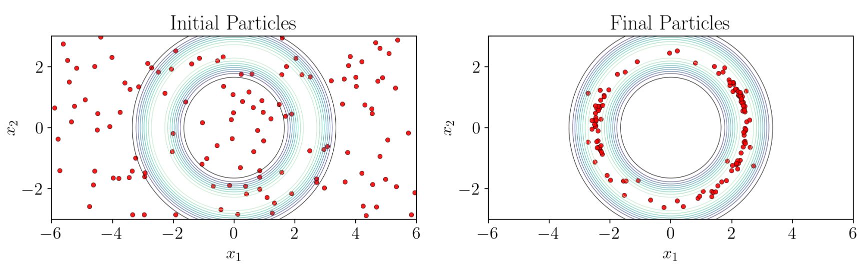
(a) Banana.



(b) Squiggly.



(c) Two moons.



(d) Ring.

Figure 3: 2D target visualization panels produced with Adapt-MR-SVGD under visualization-only settings. Each subfigure shows target-density contours with initial particles (left) and short-run final particles (right).

Method	Best KSD↓	mean logp@best↑	grad@best↓	wall@best (s)↓
Banana				
Strang-SVGD	0.114	-0.426	100	1.35
MR-SVGD	0.114	-0.392	80	1.13
Adapt-MR-SVGD	0.114	-0.393	160	0.80
SVGD	0.114	-0.391	80	0.32
Funnel				
MR-SVGD	0.256	-3.136	160	1.24
Adapt-MR-SVGD	0.357	-0.197	84	0.40
Strang-SVGD	0.369	-9.289	120	1.46
SVGD	0.377	-6.801	160	0.46
Ring				
Adapt-MR-SVGD	0.226	-0.004	238	0.82
MR-SVGD	64.35	-815.6	50	0.70
SVGD	131.6	-3753	20	0.16
Strang-SVGD	665.7	-7.92×10^5	240	2.79
Squiggly				
Adapt-MR-SVGD	0.719	-0.910	826	2.02
Strang-SVGD	22.49	-5.25×10^4	200	2.47
SVGD	22.60	-2.81×10^4	80	0.32
MR-SVGD	54.04	-422.7	100	1.36
Two moons				
Adapt-MR-SVGD	0.315	-0.009	182	0.67
MR-SVGD	134.3	-1250	40	0.60
Strang-SVGD	1892	-1.84×10^6	120	1.55
SVGD	2085	-2.58×10^6	80	0.34

Table 1: 2D particle-method summary over five seeds. For each target and method, we report medians over seeds at the best finite-KSD checkpoint. Methods are ordered by median KSD within each target block. Banana shows little separation, funnel remains nuanced, and Adapt-MR-SVGD is strongest on ring, squiggly, and two moons.

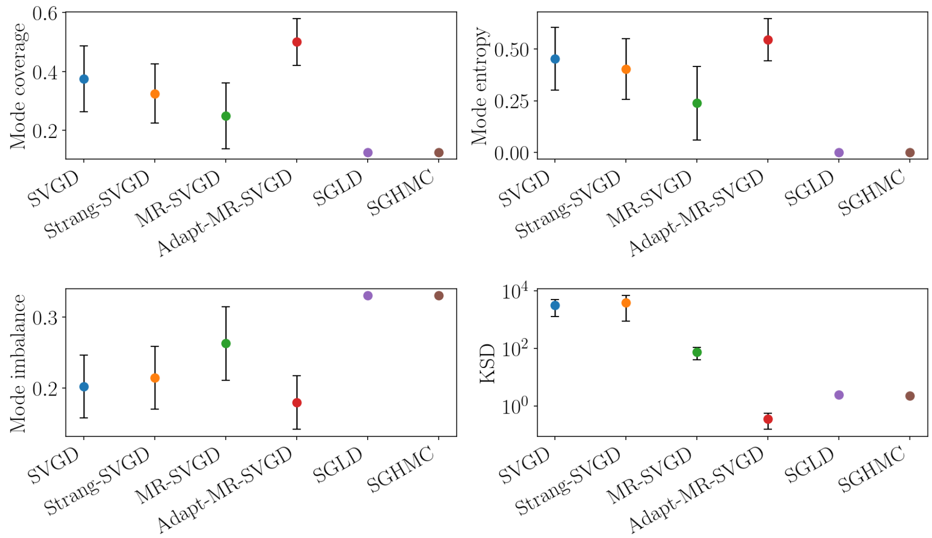
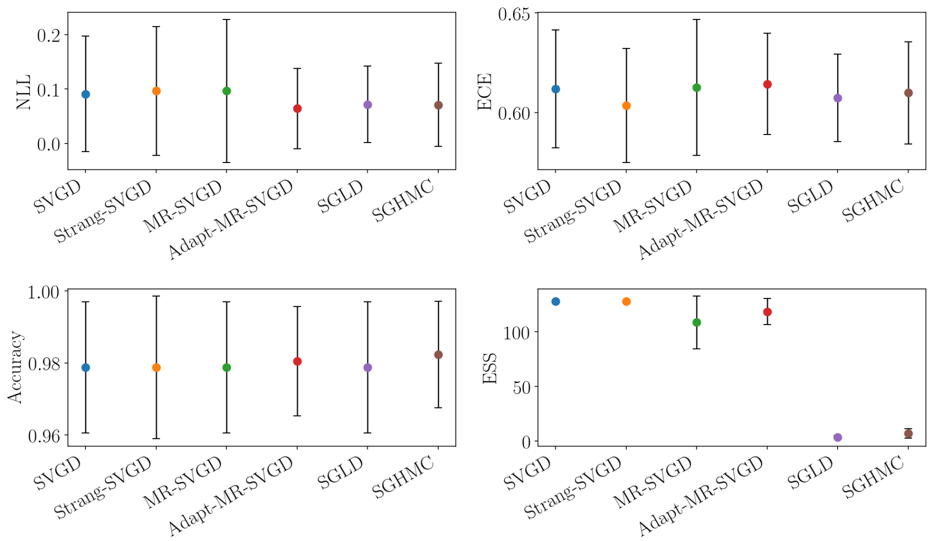
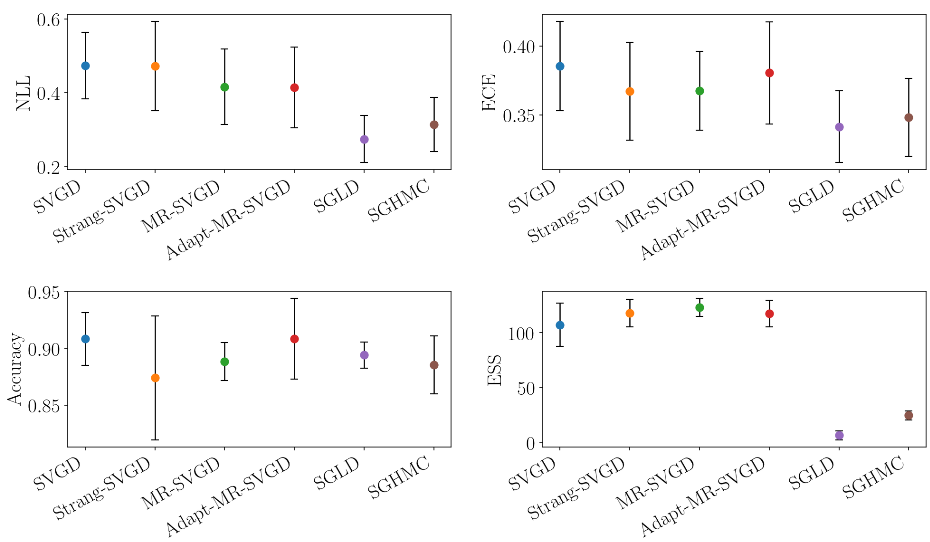


Figure 4: Mixture2D (mix8) fixed-budget final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, mode imbalance, and KSD. Higher is better for coverage and entropy, while lower is better for mode imbalance and KSD.

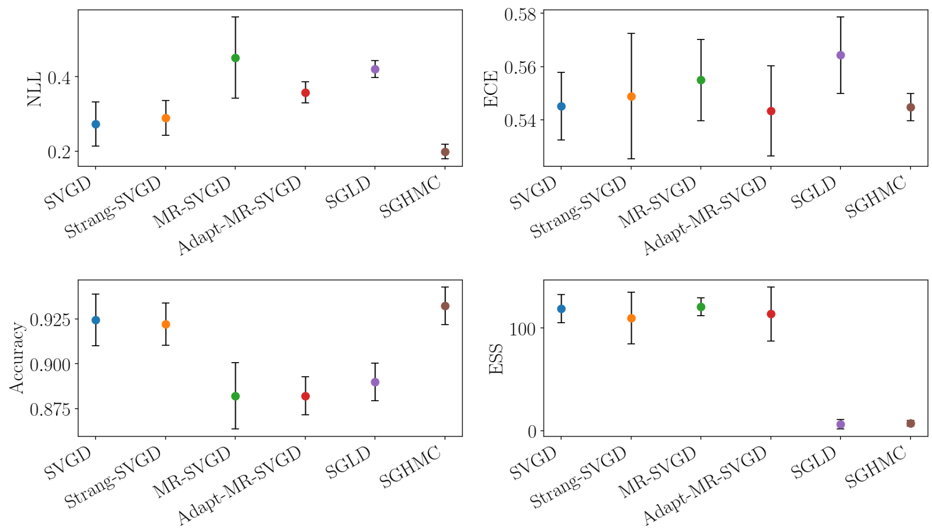


(a) breast_cancer

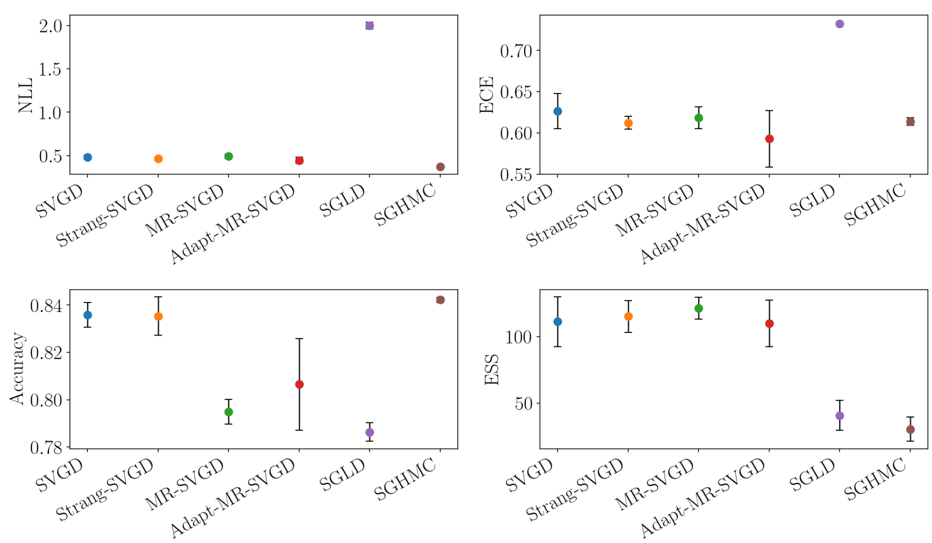


(b) ionosphere

Figure 5: UCI logistic regression summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the NLL-based early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE. Remaining datasets are shown in the continued panel.



(c) spambase

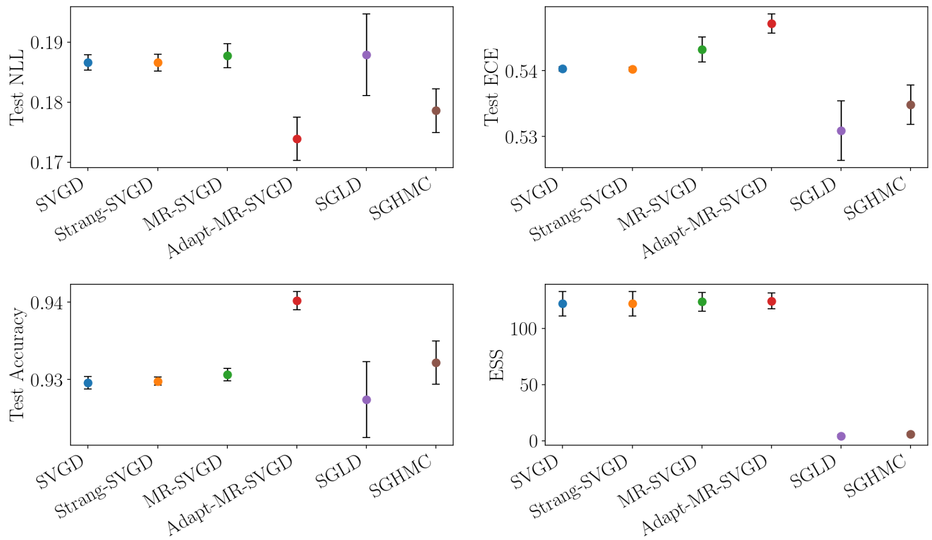


(d) a5a

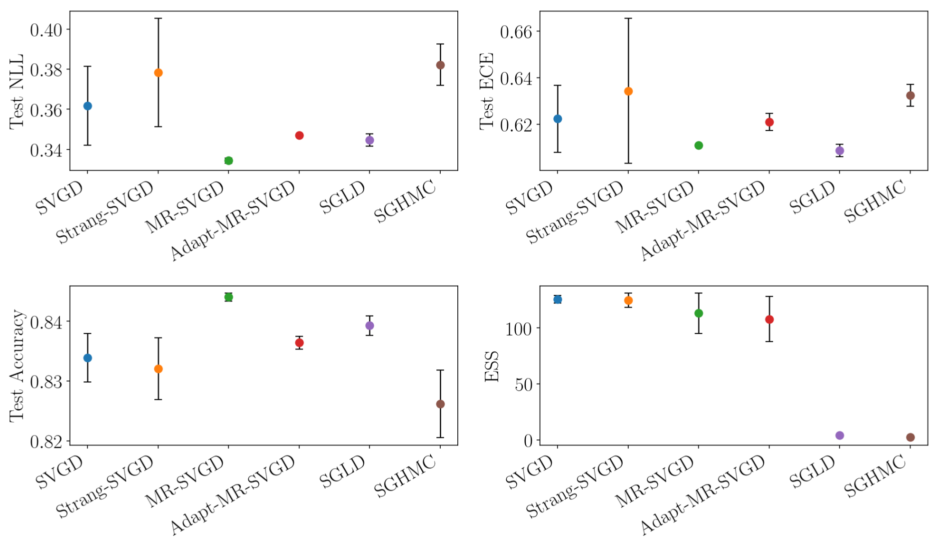
Figure 5: UCI logistic regression summary across datasets (continued).

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
breast_cancer			
Adapt-MR-SVGD	0.028	0.987	3.7
SVGD	0.091	0.979	1.4
Strang-SVGD	0.096	0.979	4.0
MR-SVGD	0.096	0.979	3.4
ionosphere			
Adapt-MR-SVGD	0.414	0.909	5.2
MR-SVGD	0.416	0.889	3.9
Strang-SVGD	0.473	0.874	3.0
SVGD	0.473	0.909	0.6
spambase			
SVGD	0.273	0.925	0.8
Strang-SVGD	0.289	0.922	3.3
Adapt-MR-SVGD	0.358	0.882	6.6
MR-SVGD	0.452	0.882	3.9
a5a			
Adapt-MR-SVGD	0.444	0.806	14.2
Strang-SVGD	0.463	0.835	2.9
SVGD	0.482	0.836	1.6
MR-SVGD	0.500	0.793	6.6

Table 2: UCI logistic-regression particle-method summary across the five-seed suite. For each dataset and method, we select the best finite-NLL checkpoint per seed and report the method-wise mean over the finite best-checkpoint values for NLL, the corresponding accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 5.



(a) spambase



(b) a5a

Figure 6: BNN summary across datasets. Each subpanel reports the best-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS, where the checkpoint is selected by the validation-NLL early-stop monitor. Higher is better for accuracy and ESS; lower is better for NLL and ECE.

Method	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
spambase			
Adapt-MR-SVGD	0.174	0.940	35.7
Strang-SVGD	0.187	0.930	51.2
SVGD	0.187	0.930	21.1
MR-SVGD	0.188	0.931	51.5
a5a			
MR-SVGD	0.334	0.844	5.3
Adapt-MR-SVGD	0.347	0.836	8.1
SVGD	0.362	0.834	17.5
Strang-SVGD	0.378	0.832	49.6

Table 3: BNN particle-method summary over five seeds. For each dataset and method, we select the best finite-validation-NLL checkpoint per seed and report the method-wise mean of the corresponding test NLL, test accuracy, and wall time. ECE is summarized in the per-dataset panels of fig. 6.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 4: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint; the table reports method-wise means of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. "Seeds with finite best" counts runs that reached at least one finite-NLL checkpoint before stopping. This table is the main hierarchical stress test: Adapt-MR-SVGD achieves the best best-NLL while remaining finite on all seeds, and MR-SVGD is the closest lower-cost baseline.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 5: HLR uniform-group summary over five seeds. As in table 4, each entry uses the best finite-NLL checkpoint per seed and reports method-wise means for NLL, accuracy at that checkpoint, and wall time.